

Labour Productivity & Comparative Advantage

ECON 308 International Economic Relations | Chapter 3: The Ricardian Model

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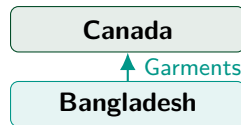
Fall 2026

Opening Puzzle: How Can Bangladesh Compete?

- Bangladesh has **lower labour productivity than Canada in every industry**: garments, electronics, agriculture, steel.
- Yet Bangladesh is a **top-five global garment exporter** (\$34 billion, 2019).
- Canada imports Bangladeshi clothing by the billions despite being more productive at *everything*.
- **Why does trade happen?**

David Ricardo's Answer (1817)

Absolute productivity does *not* determine trade. A country exports where its **opportunity cost is lowest**. This is **comparative advantage**.



Canada's Comparative Advantage:
electronics, oil, aircraft and finance

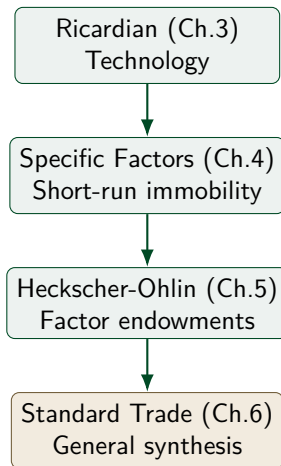
Bangladesh's Comparative Advantage:
garments

Where We Are in the Course

Chapter 3 is the foundation of all trade theory:

- **Ch. 2** (Lecture 1): *What* are the facts? Who trades with whom? The gravity model.
- **Ch. 3** (today): *Why* trade? Comparative advantage determines the pattern of trade.
- **Ch. 4**: Short run – capital is sector-specific. Trade creates winners *and* losers within a country.
- **Ch. 5**: Long run – the Heckscher-Ohlin model. Factor endowments determine trade.

Why one factor? The Ricardian model uses *labour only* to isolate technology as the source of comparative advantage. Capital and land are added in Chapters 4–5.



Learning Objectives

After this lecture you will be able to:

1. Distinguish **absolute advantage** from **comparative advantage** (CA) and explain why only CA determines trade.
2. Compute **opportunity costs** from unit labour requirements (ULRs).
3. Derive a country's linear **production possibility frontier** and autarky relative price.
4. Explain the **pattern of specialisation** and the valid range of the terms of trade.
5. Show that **both countries gain** by consuming beyond their PPFs.
6. Explain why **wages reflect productivity** and refute the pauper-labour fallacy.

Krugman, Obstfeld & Melitz, *International Economics*, 12th ed., Chapter 3.

Absolute vs. Comparative Advantage

Absolute Advantage

Country A needs *fewer labour hours* per unit: $a_{LX} < a_{LX}^*$.
A is simply more efficient.

Comparative Advantage

Country A's *opportunity cost* of X is lower:

$$\frac{a_{LX}}{a_{LY}} < \frac{a_{LX}^*}{a_{LY}^*}$$

A gives up less of Y to produce one extra unit of X.

Ricardo's insight: Even if A is better at *everything*, trade is mutually beneficial as long as opportunity costs differ. What you *give up* matters.

The key test for CA

Compare the *ratio* a_{LX}/a_{LY} across countries, not the levels. The country with the *lower ratio* has CA in X.

Cloth/Wine example:

	Home	Foreign
a_{LC}/a_{LW}	1/2	6/3 = 2
$1/2 < 2 \Rightarrow$ Home: CA in cloth .		
Reciprocal: Foreign: CA in wine .		

Opportunity Cost: The Step-by-Step Logic

OC of cloth for Home ($a_{LC} = 1$, $a_{LW} = 2$):

- 1 cloth requires $a_{LC} = 1$ labour hour.
- Those hours could produce $a_{LC}/a_{LW} = 1/2$ wine.
- OC of 1 cloth = $1/2$ wine.**

Autarky price under perfect competition:

$$\frac{P_C}{P_W} = \frac{w a_{LC}}{w a_{LW}} = \frac{a_{LC}}{a_{LW}} = \text{opportunity cost}$$

CA condition (Home has CA in cloth):

$$\underbrace{\frac{a_{LC}}{a_{LW}}}_{\text{Home OC}} < \underbrace{\frac{a_{LC}^*}{a_{LW}^*}}_{\text{Foreign OC}}$$

OC of wine for Home:

1 wine = $a_{LW} = 2$ hours, could yield $a_{LW}/a_{LC} = 2$ cloth. OC of wine = 2 cloth.

OCs are reciprocals: $OC_C \times OC_W = 1$.

If Home has CA in cloth, Foreign *must* have CA in wine.

Autarky Price = OC

Under perfect competition, relative prices equal relative unit labour costs – which equal opportunity costs.

Technology maps directly to prices:
 $P_C/P_W = a_{LC}/a_{LW}$ in autarky.

Numerical Example: Home and Foreign

Country	a_{LC}	a_{LW}	OC of cloth	OC of wine	CA
Home	1 hr	2 hrs	$\frac{1}{2}$ wine	2 cloth	Cloth
Foreign	6 hrs	3 hrs	2 wine	$\frac{1}{2}$ cloth	Wine

Absolute advantage:

- Cloth: Home (1 hr) vs. Foreign (6 hrs) $\Rightarrow$ Home $6\times$ faster.
- Wine: Home (2 hrs) vs. Foreign (3 hrs) $\Rightarrow$ Home $1.5\times$ faster.
- **Home absolutely better at both goods.**

Comparative advantage:

- OC of cloth: Home $1/2 <$ Foreign 2.
 $\Rightarrow$ **Home: CA in cloth.**
- OC of wine: Foreign $1/2 <$ Home 2.
 $\Rightarrow$ **Foreign: CA in wine.**

The Trade Prediction

Home exports cloth; Foreign exports wine – even though Home is absolutely better at both.

Trade as Indirect Production: The Key Intuition

Trade offers an alternative “production technology”:

Home can obtain wine two ways:

1. **Directly:** use $a_{LW} = 2$ hrs per unit.
2. **Indirectly:** make 1 cloth (1 hr), sell at world price p^* , buy p^* units of wine.

If $p^* > 1/2 = a_{LC}/a_{LW}$: indirect method yields *more* wine per hour.

⇒ Home always prefers cloth and trades for wine.

With $p^* = 1$:

Direct: 2 hrs → 1 wine.

Indirect: 1 hr → 1 cloth → 1 wine.

Trade halves Home's labour cost of wine.

Why Both Gain

Each country uses trade to obtain the other's good more cheaply than it could produce it.

Home's cost of wine via trade: 1 cloth/wine.

Home's cost of wine directly: 2 cloth/wine.

Foreign's cost of cloth via trade: 1 wine/cloth.

Foreign's cost of cloth directly: 2 wine/cloth.

Both gain because trade is strictly more efficient than autarky production of the imported good.

The Pauper-Labour Fallacy

The Fallacy

“Countries with cheap labour will take all our jobs. We cannot compete with \$0.57/hour wages.”

Why this is wrong:

- Low wages *reflect* low productivity. Bangladesh pays \$0.57/hr because workers produce far fewer units per hour.
- What matters is **unit labour cost** (ULC) = wage $\div$ productivity.
- Canada's CA lies in energy, aerospace, and financial services – sectors where its ULC is competitive *despite* high wages.

Unit labour cost comparison:

Garments:

Canada: $\$25/\text{hr} \div 100 \text{ units/hr} = \$0.25/\text{unit}$

BD: $\$0.57/\text{hr} \div 4 \text{ units/hr} = \$0.14/\text{unit}$

$\Rightarrow$ Bangladesh wins on garments.

Aircraft:

Canada: $\$25/\text{hr} \div 100 \text{ units/hr} = \$0.25/\text{unit}$

BD: $\$0.57/\text{hr} \div 1 \text{ unit/hr} = \$0.57/\text{unit}$

$\Rightarrow$ Canada wins on aircraft.

Key: compare unit costs, not wages.

Ricardian Verdict

High wages + high productivity = competitive. The fallacy ignores the productivity side of the equation.

The Ricardian Model: Setup

The simplest possible trade model:

- **2 countries:** Home (H) and Foreign (F*).
- **2 goods:** Cloth (C) and Wine (W).
- **1 factor:** Labour (L) only.
- Labour: **mobile** between sectors within a country; **immobile** between countries.
- **Perfect competition:** price = unit cost.
- **Constant returns to scale.**
- **Full employment.**

Key variables:

a_{LC}, a_{LW} = unit labour requirements (Home)

a_{LC}^*, a_{LW}^* = ULRs (Foreign)

L, L^* = labour endowments

w, w^* = wage rates

$p \equiv P_C/P_W$ = relative price of cloth

Under competition:

$P_C = w a_{LC}, P_W = w a_{LW}$

Autarky price: $P_C/P_W = a_{LC}/a_{LW}$

What this model cannot explain: Winners and losers within a country; role of capital or land endowments; intra-industry trade.

Home's Production Possibility Frontier

Labour constraint:

$$a_{LC} Q_C + a_{LW} Q_W = L$$

With $a_{LC} = 1$, $a_{LW} = 2$, $L = 120$:

$$Q_C + 2 Q_W = 120$$

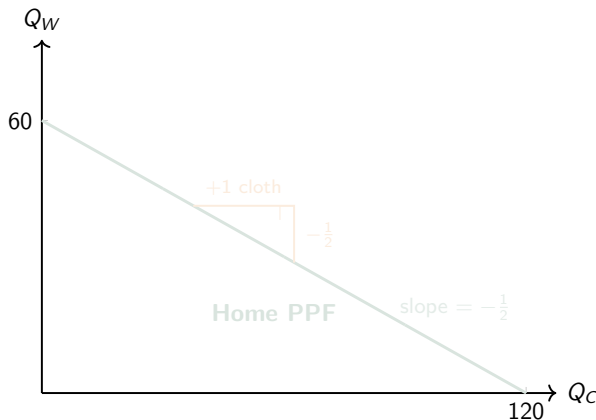
PPF intercepts:

- All cloth: $Q_C^{\max} = 120$
- All wine: $Q_W^{\max} = 60$

Slope: $-a_{LC}/a_{LW} = -\frac{1}{2}$

This slope is the **opportunity cost** of cloth. It is *constant* (linear PPF) in the Ricardian model.

Autarky price: $(P_C/P_W)^{\text{aut}} = \frac{1}{2}$



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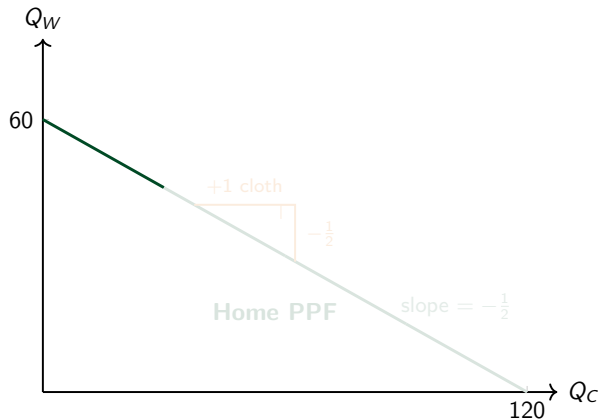
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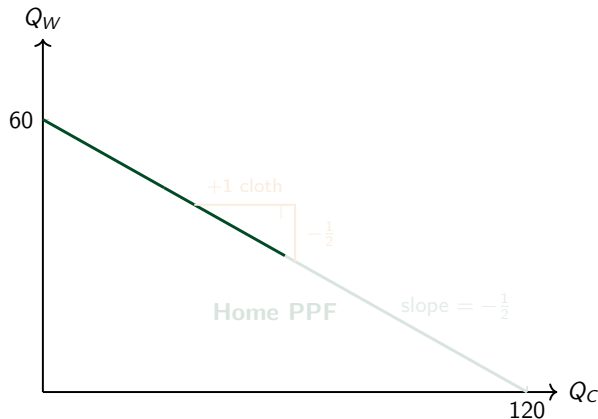
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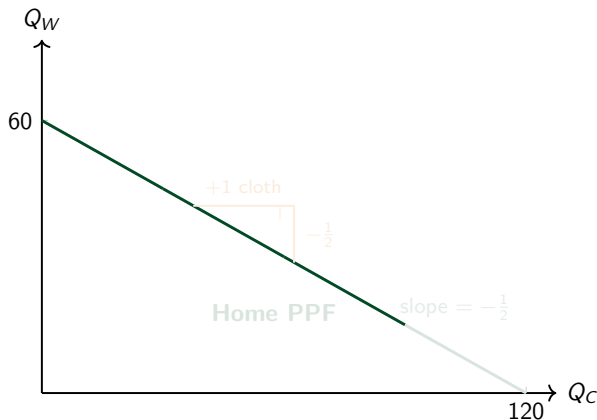
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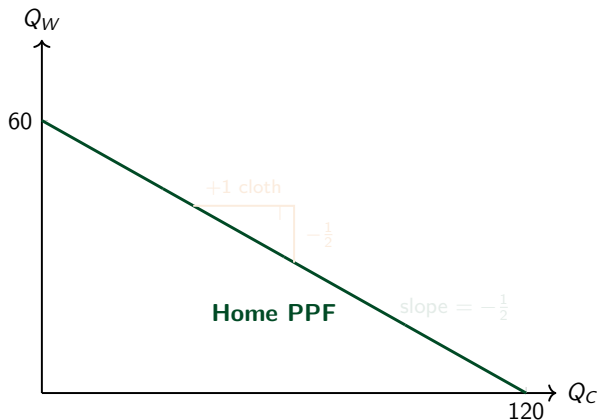
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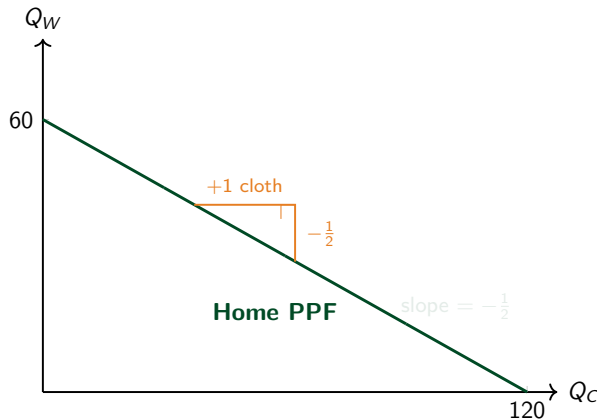
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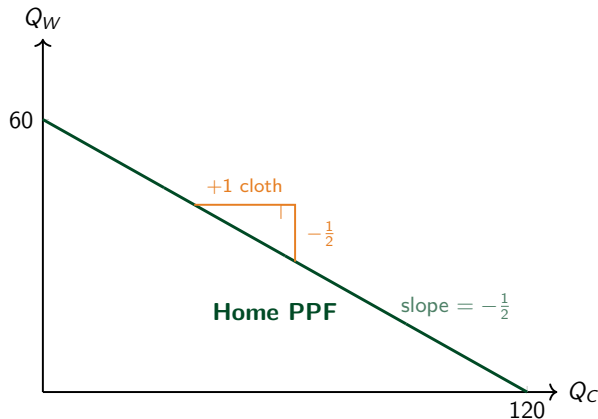
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Foreign's Production Possibility Frontier

Labour constraint:

$$a_{LC}^* Q_C^* + a_{LW}^* Q_W^* = L^*$$

With $a_{LC}^* = 6$, $a_{LW}^* = 3$, $L^* = 120$:

$$6 Q_C^* + 3 Q_W^* = 120$$

PPF intercepts:

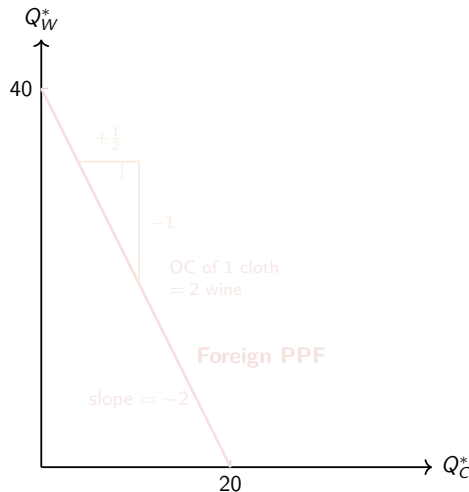
- All cloth: $Q_C^{* \max} = 20$
- All wine: $Q_W^{* \max} = 40$

Slope: $-a_{LC}^*/a_{LW}^* = -2$

Steeper than Home ($-1/2$).

Autarky price: $(P_C/P_W)^* = 2$

Foreign's cloth is more expensive (higher OC) –
this gap with Home's price of $1/2$ **drives trade**.



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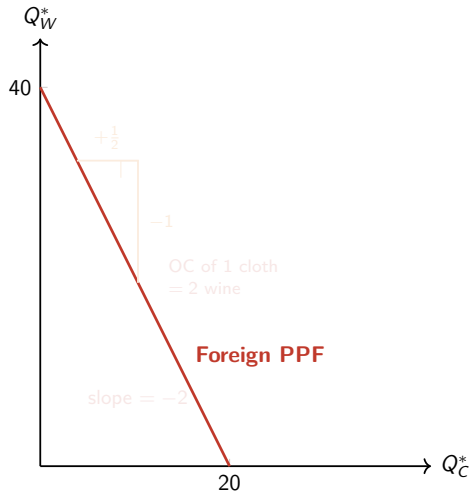
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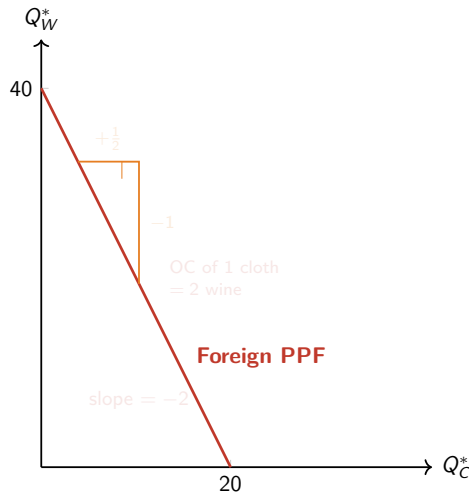
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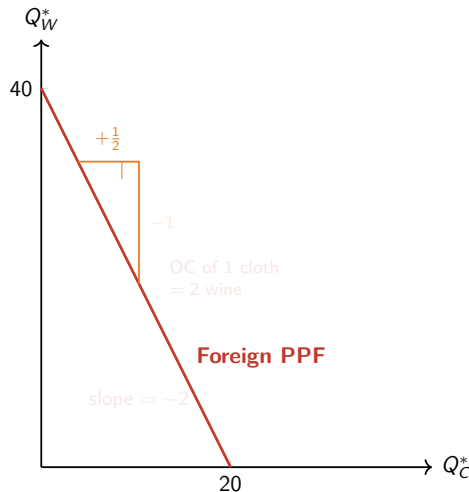
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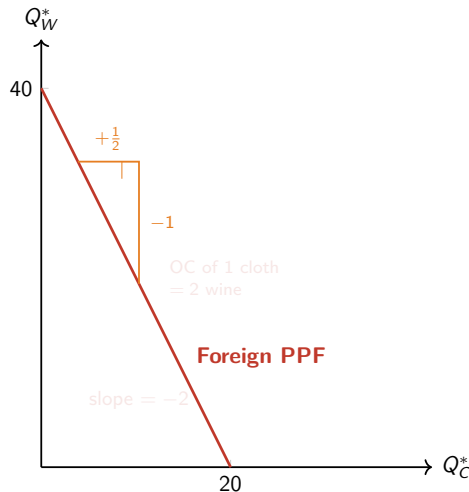
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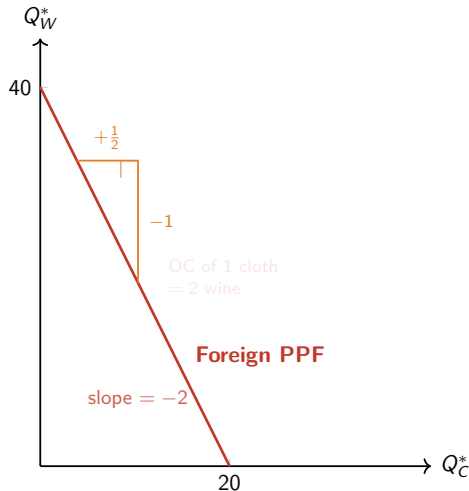
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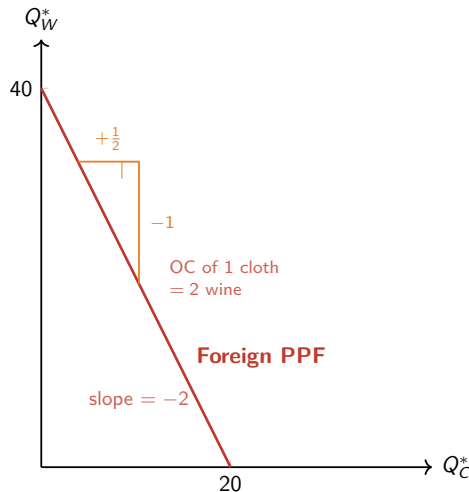
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Comparing Home and Foreign: The Price Gap

Country	a_{LC}	a_{LW}	Autarky P_C/P_W	CA in
Home	1	2	$\frac{1}{2}$	Cloth
Foreign	6	3	2	Wine

How the price gap creates trade:

1. Cloth is *relatively cheap* in Home ($P_C/P_W = 1/2$); wine is cheap in Foreign ($P_C/P_W = 2$).
2. Opening trade lets each country buy the relatively cheap good from the other.
3. Any world price p^* with $1/2 < p^* < 2$ makes **both** countries better off.

The No-Trade Trap

Without trade, each country is confined to its own PPF.

The autarky price gap ($1/2$ vs. 2) is a *mutually beneficial* trade opportunity: Home can sell cloth at a price above its OC; Foreign can sell wine at a price above its OC.

The Ricardian Trade Theorem

Ricardian Trade Theorem

Under free trade with world relative price $p^* \equiv P_C/P_W$, each country **completely specialises** in its CA good provided:

$$\frac{a_{LC}}{a_{LW}} < p^* < \frac{a_{LC}^*}{a_{LW}^*} \iff \frac{1}{2} < p^* < 2$$

Intuition:

- $p^* > 1/2$: Home earns more per hour in cloth $\Rightarrow$ all labour to cloth.
- $p^* < 2$: Foreign earns more per hour in wine $\Rightarrow$ Foreign fully in wine.
- **Home exports cloth; Foreign exports wine.**

Edge cases:

$p^* = 1/2$: Home indifferent (kink of RS).

$p^* = 2$: Foreign indifferent.

p^* outside $(1/2, 2)$: one country does not specialise completely; equilibrium is at a vertical segment of the RS curve.

World Relative Supply and Demand

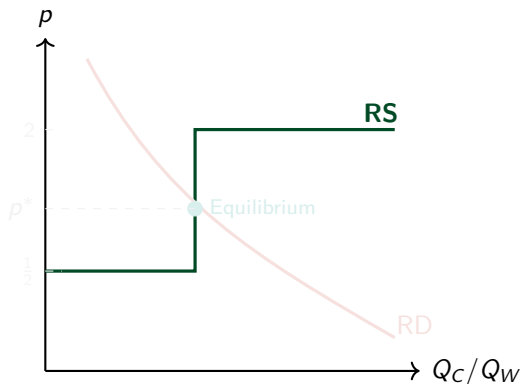
World Relative Supply (RS) = world cloth output $\div$ world wine output, as a function of p .

- $p < 1/2$: both prefer wine $\Rightarrow RS = 0$.
- $p = 1/2$: Home indifferent.
- $1/2 < p < 2$: both specialise.
- $p = 2$: Foreign indifferent; $RS \rightarrow \infty$.

World Relative Demand (RD):

downward-sloping. Higher $p \Rightarrow$ shift toward wine.

Equilibrium p^* : where RD crosses RS. If RD crosses the *inelastic segment*, both countries completely specialise.



World Relative Supply and Demand

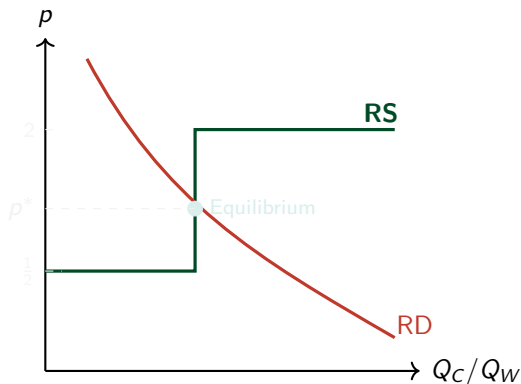
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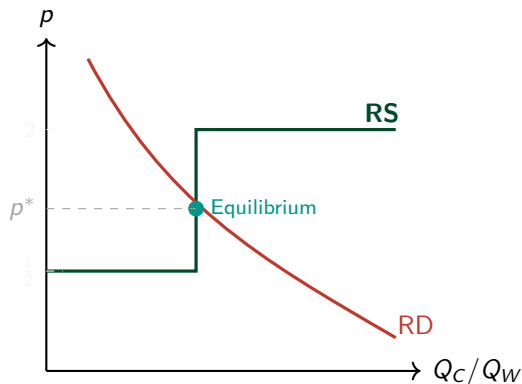
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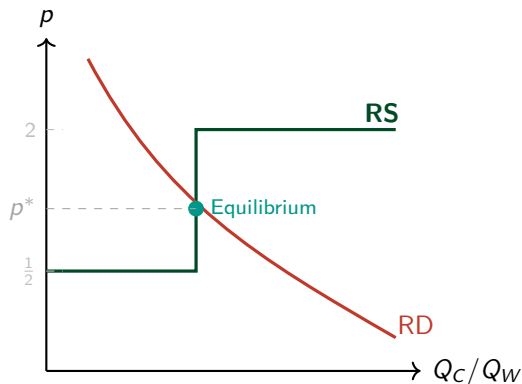
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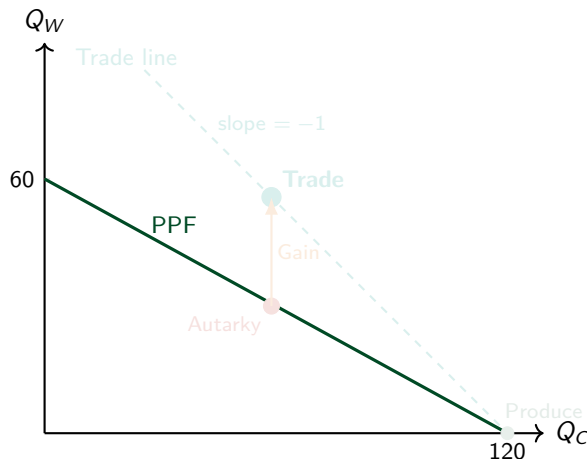
Gains from Trade: Home

Autarky: Home consumes on PPF (slope $-1/2$).

Free trade at $p^* = 1$:

1. Specialise fully in cloth: produce 120 cloth, 0 wine.
2. Sell cloth, buy wine at $p^* = 1$.
3. Budget line (trade line) through production point (120, 0) with slope -1 .
4. Home reaches a higher indifference curve **above** the PPF.

The gain: domestic cost of wine = 2 cloth/wine; trade cost = 1 cloth/wine.
Trade saves 1 unit of cloth per wine imported.



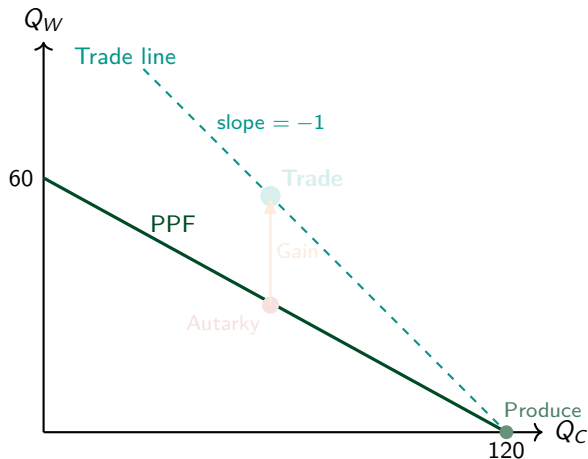
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2. Sell cloth, buy wine at $p^* = 1$.
3. Budget line (trade line) through production point (120, 0) with slope -1 .
4. Home reaches a higher indifference curve **above** the PPF.

The gain: domestic cost of wine = 2 cloth/wine; trade cost = 1 cloth/wine.
Trade saves 1 unit of cloth per wine imported.



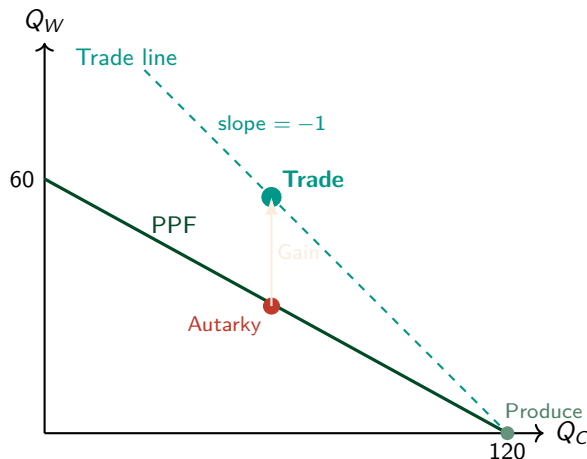
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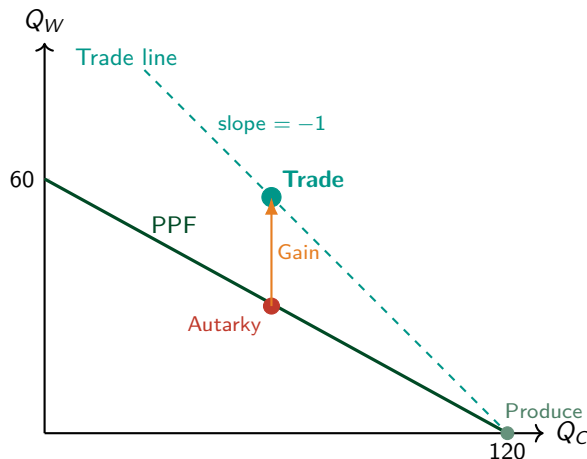
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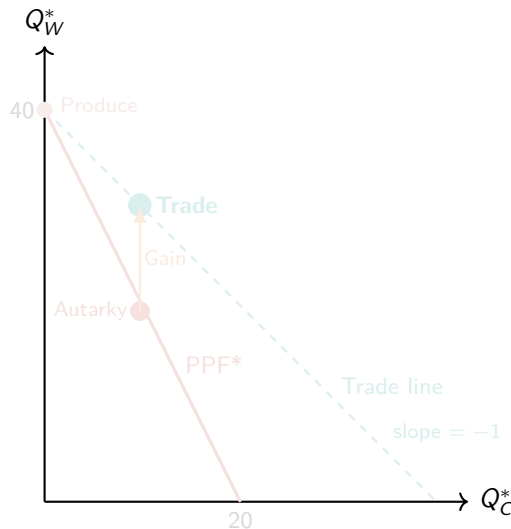
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Foreign autarky: PPF slope = -2 .

Free trade at $p^* = 1$:

1. Specialise fully in wine: produce 40 wine, 0 cloth.
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3. $p^* = 1 < 2$: cloth cheaper via trade than domestic production.
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Trade halves Foreign's cost of obtaining cloth.



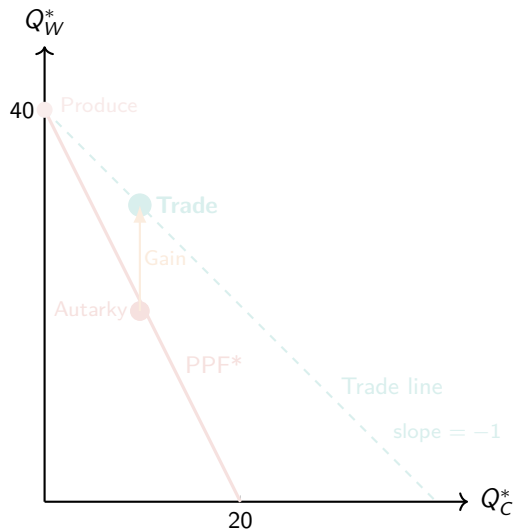
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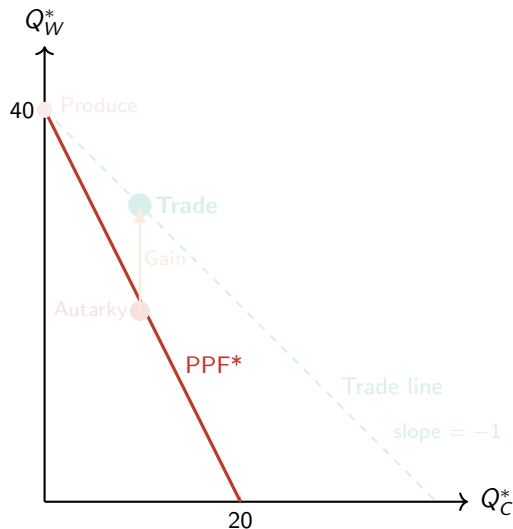
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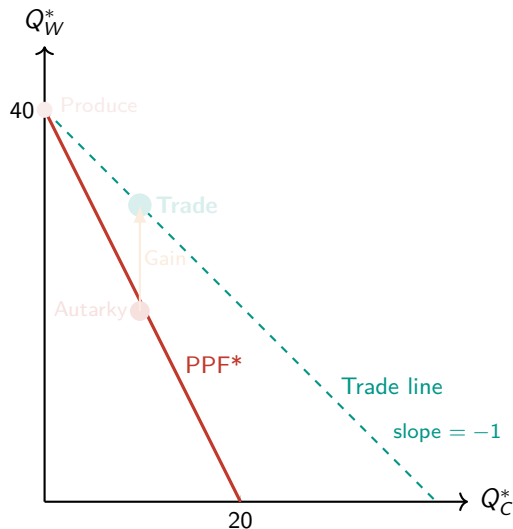
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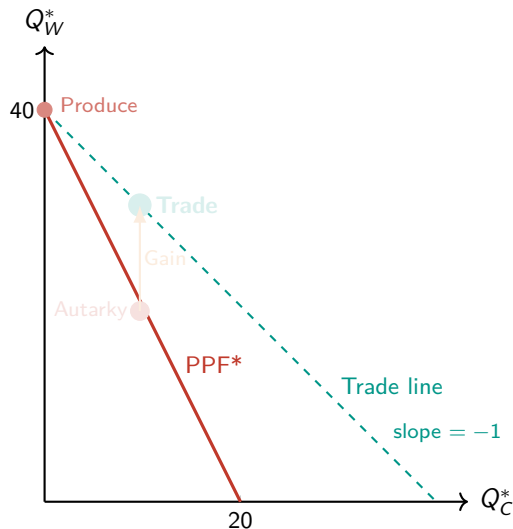
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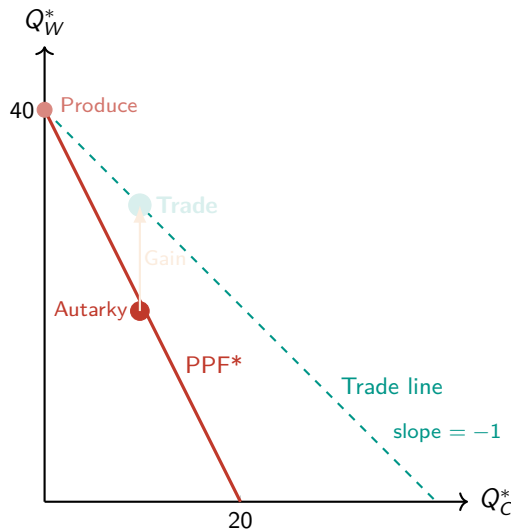
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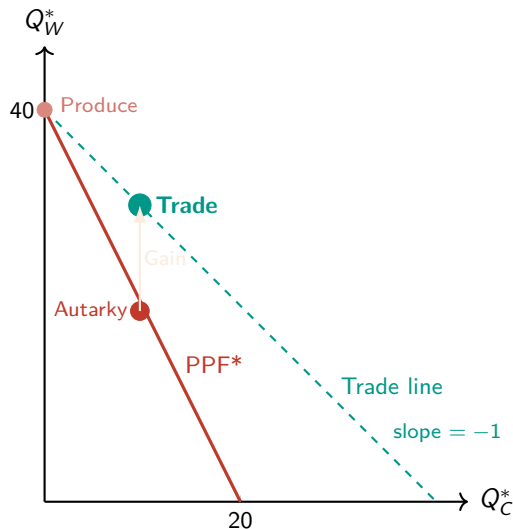
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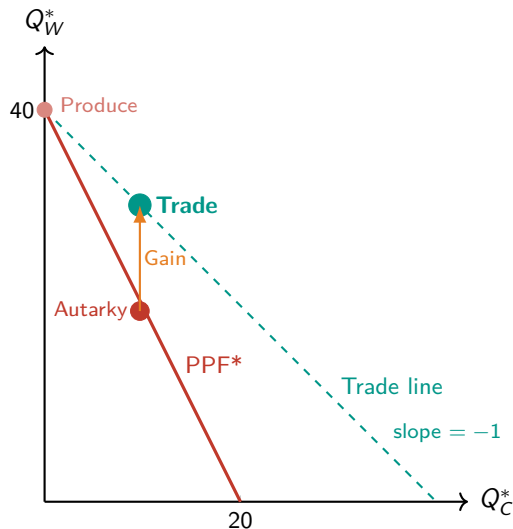
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The Terms of Trade: Definition and Welfare

Terms of Trade (ToT)

ToT for Home = export price $\div$ import price
 $= P_C/P_W = p^*$.

Mutual gain requires: $\frac{a_{LC}}{a_{LW}} < p^* < \frac{a_{LC}^*}{a_{LW}^*}$

Effect of ToT on welfare:

- **Higher p^* (toward 2):** Home gains more; each cloth exported buys more wine.
- **Lower p^* (toward 1/2):** Foreign gains more.
- Both *always* gain relative to autarky as long as $1/2 < p^* < 2$.

Equilibrium p^* : RS = RD.

Canada's Terms of Trade

Canada exports energy, agriculture, aircraft; imports manufactures, electronics.

2002–2014: Rising oil prices raised Canada's ToT sharply. Real national income rose $\approx 15\%$ more than real GDP – a pure ToT gain.

After 2014: Oil price collapse worsened ToT. Alberta entered recession; Bank of Canada cut rates in 2015–16.

Ricardian lesson: exporters of a rising-price good gain.

Wages, Productivity, and Trade

Home's wage (specialises in cloth):

$$w = \frac{P_C}{a_{LC}} = \frac{p^* P_W}{a_{LC}}, \quad w^* = \frac{P_W}{a_{LW}^*}$$

Relative wage ratio:

$$\frac{w}{w^*} = p^* \cdot \frac{a_{LW}^*}{a_{LC}}$$

With $p^* = 1$, $a_{LC} = 1$, $a_{LW}^* = 3$: $w/w^* = 3$.

Home earns 3× Foreign wages – because it is 3–6× more productive.

Unit labour costs:

- Cloth: Home = $3w^*$; Foreign = $6w^*$. Home cheaper. ✓
- Wine: Home = $6w^*$; Foreign = $3w^*$. Foreign cheaper. ✓

Wage Result

Higher productivity $\Rightarrow$ higher wages.
But high wages do *not* prevent trade.

Home's wage advantage is exactly offset by its productivity advantage in cloth.

Each country is competitive in its CA sector *regardless* of the absolute wage level.

Relative-wage bounds:

$$\frac{a_{LC}^*}{a_{LC}} > \frac{w}{w^*} > \frac{a_{LW}^*}{a_{LW}}$$

In example: $6 > w/w^* > 3/2$.

At $p^* = 1$: $w/w^* = 3$ (✓ inside bounds).

Do Wages Reflect Productivity? The Ricardian Prediction

Ricardian model predicts: $w \propto$ labour productivity.

Country	Wage (USD/hr)	Productivity (USA = 100)	Unit labour cost
USA	35.0	100	35.0¢/unit
Germany	43.0	108	39.8¢/unit
South Korea	22.0	67	32.8¢/unit
Mexico	5.0	28	17.9¢/unit
Bangladesh	0.6	4	15.0¢/unit

BLS, ILO, World Bank (2019)

Key findings:

- High-productivity countries have high wages: Ricardian prediction **confirmed**.
- Unit labour cost varies less than wages – productivity differences offset wage differences.
- Bangladesh's low wages reflect low productivity, not a threat to Canadian workers in high-productivity sectors.

Verdict

Trade with low-wage countries does *not* impoverish high-wage countries – because low wages come with low productivity.

Bangladesh vs. China: CA in Action

Both export garments. How?

- China has higher absolute productivity in *both* garments and electronics.
- But China's **OC of garments is rising**: its CA shifts toward electronics, AI, and capital-intensive goods.
- Bangladesh's OC of garments remains low – it lacks a competitive electronics sector.
- China moves up the value chain; Bangladesh stays in garments.

Confirmed

Each country exports where its *relative* cost is lowest, not where its absolute cost is lowest.

	Bangladesh	China
Mfg wage (USD/hr)	0.57	4.99
Garment exports	\$34B	\$185B
Electronics exp.	\$0.4B	\$842B
Garments/Elect.	85×	0.2×

ITC, World Bank (2019)

Worked Example: Wine and Cheese

Worked Example

Two countries, **Home** and **Foreign**; two goods, **Wine (W)** and **Cheese (C)**. $L = L^* = 120$ hours of labour each.

	Wine (hr/unit)	Cheese (hr/unit)
Home	$a_{LW} = 2$	$a_{LC} = 1$
Foreign	$a_{LW}^* = 6$	$a_{LC}^* = 2$

Questions:

- Max Wine and Cheese for each country.
- OC of Wine in terms of Cheese (each country).
- Which country has CA in Wine? In Cheese?
- At $P_W/P_C = 2.5$: who exports what?

Steps:

1. Max $Q = L/a_L$.
2. OC of Wine = a_{LW}/a_{LC} .
3. Lower OC $\Rightarrow$ CA in Wine.
4. Compare p^* to autarky price:
 $p^* > \text{autarky} \Rightarrow \text{export Wine}.$

Worked Example: Solution

(a) Max production ($L = 120$):

- Home: $Q_W^{\max} = 60$; $Q_C^{\max} = 120$.
- Foreign: $Q_W^{*\max} = 20$; $Q_C^{*\max} = 60$.

(b) OC of Wine:

- Home: $a_{LW}/a_{LC} = 2/1 = 2$ Cheese.
- Foreign: $6/2 = 3$ Cheese.

(c) CA: $2 < 3 \Rightarrow$ Home: Wine; Foreign: Cheese.

(d) At $P_W/P_C = 2.5$:

- Home autarky price = $2 < 2.5$: Wine is profitable. $\Rightarrow$ **Home exports Wine.**
- Foreign autarky price = $3 > 2.5$: Wine cheaper via trade. $\Rightarrow$ **Foreign exports Cheese.**

Key Check

Valid range for P_W/P_C : $(2, 3)$.

$p^* = 2.5$ lies in this range. ✓

$2 < 2.5$: Home exports Wine. ✓

$3 > 2.5$: Foreign exports Cheese. ✓

Both specialise completely; both gain.

Mutual-gain check:

Home gains if $p^* > 2$. ✓

Foreign gains if $p^* < 3$. ✓

$p^* = 2.5$ satisfies both. ✓

Limitations and What Comes Next

What the Ricardian model does well:

- Explains the *direction* of trade (CA).
- Shows *mutual gains* for all countries.
- Connects productivity to wages (empirically supported).
- Refutes the pauper-labour fallacy.
- Generates testable predictions (MacDougall).

What it misses:

- Complete specialisation (never fully observed).
- No capital or land (factor endowments ignored).
- No losers within a country.
- No economies of scale; no intra-industry trade.

Building Up the Model

- **Ch. 4** (Specific Factors): Add capital and land. Trade creates winners and losers *within* a country.
- **Ch. 5** (H-O): Long run – both factors mobile. Factor endowments determine CA.
- **Ch. 7** (EoS): Increasing returns explain intra-industry trade.

Ricardian $\subset$ Specific Factors $\subset$ H-O

Each model is a special case of the next.

Key Points

1. **Comparative advantage**, not absolute advantage, determines trade. Export where OC is *lowest*.
2. OC = ratio of ULRs: a_{LC}/a_{LW} .
3. **Ricardian PPF is linear** (constant OC); slope $= -a_{LC}/a_{LW}$ = autarky price.
4. Free trade $\Rightarrow$ **complete specialisation** in CA good.
5. Both countries **consume beyond PPFs** – mutual gains.
6. **ToT** must lie strictly between the two autarky prices.
7. **Wages track productivity**: high wages reflect high productivity; all countries gain.
8. **Pauper-labour fallacy** is wrong: unit labour cost, not wages alone, matters.

Next Lecture: Chapter 4

Specific Factors and Income Distribution

What happens in the short run when capital is locked into specific sectors?

Who wins and who loses from trade when factor mobility is limited?

Why do some workers and firms lobby against free trade even when the nation gains?

Key Readings

- Ricardo, D. (1817), *Principles of Political Economy*, Ch. 7.
- MacDougall (1951), "British and American Exports," *Economic Journal*.
- Balassa (1963), *Review of Economics and Statistics*.
- Golub & Hsieh (2000), *Review of International Economics*.